

DATA ANALYST

Assignment -1

BY

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SECTION 1 : CORE CONCEPTS

Q1. Definitions

1. Data?

Data is **raw facts or values** collected from real life. By itself, data does not give meaning

Example : Marks = 60, 70, 80

2. Data Analytics?

Data analytics means **studying data to understand what is happening and why it is happening.**

Example : Analyzing marks to find who is weak and who is strong.

3. Difference between Data and Information

Data	Information
Data is raw details	Information is meaningful result after analysis
Example: 500, 700	Sales are increasing

Q2. Why companies use Data Analytics

- To take **better decisions**
- To **understand customers**
- To **increase profit**
- To **find problems early**
- To **plan for future**

Example:Netflix uses data to decide which movies people like.

SECTION 2: Types of Data

Q3. Identify type and nature of data

Scenario	Data Type	Nature
a) Customer age	Structured	Quantitative
b) Netflix movie review text	Unstructured	Qualitative
c) Bank transaction table	Structured	Quantitative
d) Uber GPS coordinates	Semi-Structured	Quantitative
e) Amazon ratings (1–5)	Structured	Quantitative

Q4. Convert unstructured data into structured form

Given text : The delivery was late and customer support was poor

feedback_text	sentiment	issue_type
Delivery was late and support was poor	Negative	Delivery & Support

SECTION 3: Analytics Types Mapping

Q5. Match business question with analytics type

Business Question	Analytics Type
Why are users uninstalling our app?	Diagnostic Analytics
How many users logged in yesterday?	Descriptive Analytics
What will next month's revenue be?	Predictive Analytics
Which action should reduce churn?	Prescriptive Analytics

Q7. Analytics use cases for Netflix ?

1.Descriptive Analytics

Problem: What content is most watched?
Data: Viewing history
Output: Top 10 popular shows
Business value: Content planning

2.Predictive Analytics

Problem: Which users may cancel subscription?
Data: Watch frequency, inactivity
Output: Churn prediction list
Business value: Retention strategy

2.Diagnostic Analytics

Problem: Why did viewership drop?
Data: User activity logs
Output: Drop after price increase
Business value: Reduce churn

3.Prescriptive Analytics

Problem: How to reduce churn?
Data: Churn prediction + offers
Output: Personalized discounts
Business value: Increased revenue

SECTION 5: Analyst Thinking Test

Q8. Revenue is falling – Analyst approach

1. Questions to ask

Which product revenue is falling?
From when did it start?
Is customer count reducing?

2. Data to request

Sales data
Customer data
Marketing spend
Product performance

3. Metrics to analyze

Revenue growth
Customer churn rate
Conversion rate
Average order value

SECTION 6: Executive Summary

- Data analytics helps to understand business data.
- It helps companies take better decisions.
- It improves customer experience.
- It helps to plan future actions.
- It supports business growth.